Ontology Unpacking and Semantic Alignment for Recommender Systems

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Abstract: Strategic sales recommendation has become increasingly important for organizations seeking to maximize profitability through actions such as discount prioritization, bundle construction, and cross-selling opportunities. Although traditional recommender systems are effective at identifying user-item correlations, strategic sales recommendation requires additional semantic and business reasoning capabilities capable of supporting broader commercial decision-making processes. In this paper, we propose a knowledge-driven ontology-guided methodology for recommendations that combines semantic onboarding, ontological unpacking, semantic alignment, and logic-based reasoning. The approach transforms transactional data, product knowledge, and explicit business sales criteria into formal executable representations, enabling the construction of a semantically grounded knowledge base capable of supporting commercial inference. By modeling business relations as ontological entities and leveraging executable logic, the system generates strategic recommendations supported by explicit inference and traceable explanations. A comparative evaluation against a Knowledge Graph-based baseline highlights the fundamental distinction between correlation-driven recommendation and ontology-grounded strategic reasoning. While existing approaches explain product associations, the proposed framework explains why specific sales strategies hold, enabling more robust, actionable, and semantically justified commercial recommendations.

1 Introduction

In highly competitive retail and e-commerce environments, organizations continuously seek recommendation strategies capable of increasing revenue, improving product turnover, and maximizing overall commercial profitability. Such strategies include identifying products suitable for bundle discounts, determining which product categories should be prioritized in promotional campaigns, and recommending personalized commercial opportunities for specific customers. In this work, we adopt a broader view of recommender systems, treating strategic recommendation not merely as a predictive relevance task, but as a semantically grounded decision-support problem in which recommendations must satisfy explicit business objectives and organizational constraints.

Traditional recommender systems have historically focused on collaborative filtering, content-based similarity, and more recently knowledge-aware recommendation architectures capable of leveraging structured relational information. In particular, Knowledge Graph (KG)-based recommenders have substantially improved recommendation explainability by exposing entity relations, graph connectivity patterns, and interpretable association paths. More recently, Large Language Models (LLMs) have further enhanced recommendation systems by generating natural-language explanations and supporting richer interactive recommendation scenarios. Despite these advances, existing architectures predominantly explain why products appear associated according to observed interaction patterns, graph connectivity, or learned relational representations.

This limitation becomes particularly relevant in strategic recommendation scenarios, where the relevant decision question is not merely whether entities are behaviorally related, but whether the available organizational evidence satisfies the semantic conditions required for a specific business action. For example, recommending a bundle discount is not simply a matter of identifying co-purchased products, but of determining whether those products satisfy explicit commercial criteria such as complementarity, ecosystem affinity, or strategic promotional suitability. Similarly, prioritizing product categories for discount campaigns requires aggregating heterogeneous business evidence under explicit decision criteria rather than relying solely on behavioral proximity or statistical association. In such contexts, explainability requires more than exposing relational paths; it requires demonstrating why a recommendation proposition is semantically valid with respect to explicit business intent.

To address this limitation, we propose an ontology-guided recommendation framework that combines Semantic Onboarding, Ontological Unpacking, Semantic Bridging, and executable logic-based reasoning to support strategic recommendation generation. Rather than treating recommendation as a graph traversal or probabilistic ranking problem, the proposed approach transforms both organizational data and business sales intent into formal executable knowledge structures. Transactional evidence is semantically onboarded into an ontologically grounded knowledge model, while business objectives are formally unpacked into executable recommendation criteria. These two semantic spaces are then aligned through a Semantic Bridging layer that performs explicit logical satisfiability checks over organizational truthmakers, enabling recommendations to emerge as formally justified strategic conclusions rather than probabilistic associations.

This perspective also fundamentally changes the role of explainability in recommendation systems. In conventional explainable recommenders, explanations are often post-hoc rationalizations derived from graph connectivity or learned correlations. In contrast, in the proposed framework, explanations emerge directly from executable proof traces generated during logical inference, ensuring that recommendation justifications remain semantically grounded, traceable, and structurally aligned with business decision criteria. In this sense, while existing systems explain why products appear related, the proposed framework explains why a strategic recommendation should hold.

The main contributions of this work are threefold. First, we propose an ontology-guided methodology for strategic recommendation that explicitly aligns business intent with organizational evidence through semantic bridging. Second, we implement an executable neuro-symbolic recommendation architecture combining ontological formalization, semantic alignment, and Prolog-based logical reasoning for recommendation generation and explanation. Third, we provide a comparative evaluation against a Knowledge Graph + LLM baseline, demonstrating improvements in logical soundness, strategic recommendation quality, explainability, and computational scalability.

The remainder of this paper is organized as follows. Section 2 introduces the ontological foundations underlying our approach. Section 3 reviews the related literature on recommendation systems, explainable recommendation, and structured reasoning approaches. Section 4 presents the proposed methodology, while Section 5 details the system architecture and implementation aspects. Section 6 presents the experimental evaluation and comparative analysis. Finally, Section 7 concludes the paper and outlines future research directions.

2 Background

Traditional recommender systems were initially dominated by collaborative filtering and content-based approaches, in which recommendations are inferred from user-item similarity patterns or descriptive item attributes. Although effective in many practical settings, these models often suffer from sparsity, cold-start limitations, and low interpretability due to their dependence on latent statistical correlations. As recommender systems became increasingly embedded in business decision processes, the need for richer semantic context and more explainable recommendation rationales led to the emergence of *knowledge-aware recommender systems* (Wang et al., 2019).

In this context, KGs became one of the dominant representational paradigms because they allow heterogeneous entities to be connected through typed relational structures. According to (Huang et al., 2022), KGs enable recommender systems to capture high-order connectivity patterns and side information that are inaccessible to purely transactional recommendation models. More recently, graph neural and path-based architectures have further reinforced this trend by exploiting graph propagation and relation-aware embeddings to improve recommendation robustness and explainability (Wang et al., 2019). More recently, graph neural and path-based architectures have further reinforced this trend by exploiting graph propagation, neighborhood aggregation, and relation-aware embeddings to improve recommendation robustness and contextual explainability (Gao et al., 2022).

Despite these advances, the semantic interpretation of KG-based recommendation remains predominantly association-driven. Most systems treat graph relations as typed edges whose utility lies in propagating signals, identifying paths, or extracting neighborhood correlations. As a consequence, recommendation explanations are usually generated from graph connectivity rather than from formally grounded business semantics. This limitation becomes particularly relevant in strategic sales recommendation scenarios, where the central question is not only whether two entities are connected in the graph, but whether the observed business evidence satisfies an intended commercial objective. This distinction motivates the ontology-guided semantic grounding adopted in this work.

The Unified Foundational Ontology (UFO) was

developed to provide a rigorous ontological basis for conceptual modeling and information systems engineering. Unlike lightweight taxonomies, UFO offers a richer distinction between entities, relations, events, and intentional structures. UFO is organized into three complementary modules: UFO-A, devoted to structural entities that persist in time; UFO-B, devoted to temporally unfolding events and processes; and UFO-C, devoted to social and intentional aspects of reality (Guizzardi et al., 2015).

Within UFO-A, a central concern is the principle of identity of domain entities, a view influenced by OntoClean and by the philosophical contributions of (Strawson, 1959) and (Lowe, 2001). In this layer, *Kinds* provide rigid identity criteria to their instances, while UFO distinguishes substantials as *Functional Complexes*, i.e., functionally organized individuals, and *Collectives*, i.e., individuals formed by homogeneous members (Guizzardi and Wagner, 2010). UFO-A also introduces the distinction between *internal relations*, which derive from intrinsic properties, and *material relations*, which require a mediating ontological entity called a *relator* (Guizzardi, 2005). Relators ground the existence of externally valid relations and are existentially dependent on the entities they connect, a dependence captured by the notion of *foundation* (Guizzardi and Wagner, 2012).

UFO-B complements this structural perspective by modeling *perdurants*, i.e., events and processes that unfold temporally (Guizzardi et al., 2015). This temporal view is particularly relevant because many material relations emerge through events, such as purchase situations that instantiate user-product commercial interactions. Such ontological grounding naturally connects to the notion of *truthmakers*: according to (Guarino and Guizzardi, 2015), a truthmaker is an entity whose existence makes a proposition true. In this sense, relators function as truthmakers for material business relations. Finally, UFO-C introduces *Agents* and *Roles*, allowing entities to participate in intentional and socially contextualized situations (Guizzardi, 2008). Together, these three layers provide the ontological basis for treating recommendation evidence not as simple observed associations, but as semantically grounded commercial facts.

3 Related Work

Recommender systems have evolved substantially from traditional collaborative filtering and content-based approaches toward architectures capable of exploiting richer structured knowledge and more sophisticated reasoning mechanisms. In particular, Knowledge Graph (KG)-based recommender systems have emerged as a prominent paradigm due to their ability to explicitly represent heterogeneous entities and typed relationships, enabling the incorporation of contextual and semantic signals beyond pure behavioral similarity. According to (Huang et al., 2022), KGs allow recommendation models to integrate heterogeneous knowledge sources and capture high-order relational dependencies that are inaccessible to purely interaction-driven approaches. Architectures such as KGAT (Wang et al., 2019) further advanced this direction by leveraging graph attention mechanisms to propagate recommendation signals across relational structures, while surveys such as (Gao et al., 2022) emphasize the broader role of graph neural architectures in recommendation tasks.

Beyond predictive performance, explainability has become a central concern in recommendation research. Several approaches attempt to generate explanations by exposing relational paths, connectivity structures, or neighborhood dependencies within knowledge graphs. For instance, (Xie et al., 2021) propose an explainable recommendation framework based on graph path optimization, while (Tiwary et al., 2024) review explainable recommender systems that rely on knowledge graphs and reinforcement learning to expose interpretable recommendation rationales. More recent approaches attempt to strengthen explanation quality by introducing explicit rationalization mechanisms. (Yang et al., 2023) propose a self-supervised framework that assigns rational scores to knowledge graph triplets in order to identify explanatory evidence, while (Ren et al., 2023) explore explicit knowledge graph reasoning for conversational recommendation through reasoning chains extracted from graph connectivity.

At the same time, Large Language Models (LLMs) have increasingly been incorporated into recommendation architectures, particularly to improve interaction quality, natural-language explanation, and reasoning flexibility. Surveys such as (Lin et al., 2025) highlight the growing role of LLMs in recommendation pipelines, including graph-augmented recommendation and explanation generation. These approaches significantly improve human interpretability by producing fluent textual justifications and enabling more interactive recommendation scenarios. Complementary efforts have also explored combining structured knowledge representations with LLM prompting strategies to guide recommendation generation from explicit relational evidence (Seabra et al., 2025). However, in most of these architectures, structured knowledge primarily serves as contextual input to guide probabilistic inference or language genera-

tion rather than as an executable formal substrate for recommendation validation.

A related line of research explores neural-symbolic and logic-aware recommendation paradigms, where symbolic reasoning is incorporated to improve explainability, consistency, or reasoning robustness. For example, (Ma et al., 2021) propose a neuro-symbolic recommendation architecture that combines neural recommendation models with symbolic reasoning constraints. Such efforts represent an important step toward recommendation systems capable of moving beyond purely statistical association learning. Nevertheless, even in these architectures, symbolic reasoning is often introduced as an auxiliary constraint over predictive models rather than as the primary mechanism through which recommendation validity is established with respect to explicit business intent.

Despite these advances, an important conceptual gap remains. Existing KG-based, explainable, and LLM-enhanced recommender systems primarily explain why entities appear associated according to observed interaction patterns, graph connectivity, or learned relational representations. Even when reasoning chains are exposed, recommendation validity remains grounded in association structures rather than in explicit semantic satisfiability conditions derived from organizational objectives. In strategic decision-support scenarios, however, the relevant question is not merely whether two products are behaviorally connected, but whether the available organizational evidence satisfies the semantic conditions required for a specific business action, such as discount prioritization, bundle recommendation, or commercial opportunity targeting.

This distinction motivates the ontology-guided approach proposed in this work. Rather than treating recommendation as a graph traversal or probabilistic association task, we frame strategic recommendation as an executable semantic reasoning problem in which business intent is explicitly formalized, aligned with organizational truthmakers, and validated through logical inference. In this formulation, recommendations do not emerge merely from connectivity patterns or learned correlations, but from the formal satisfaction of business conditions grounded in organizational evidence. This enables the recommendation process to operate not only as a predictive mechanism, but as a decision support capable of assessing whether a commercial strategy is semantically justified. In this sense, the proposed framework transforms business semantics into an executable substrate for justified strategic recommendations.

4 Methodology

The proposed methodology establishes a formal pipeline to bridge the gap between business intent and organization knowledge to support decision-making in strategic sales. As illustrated in figure 1, our methodology is structured as a multi-stage pipeline designed to transition from raw information to ontological unpacked knowledge. The pipeline is divided into three self-contained stages: Semantic Onboarding, Ontological Unpacking, and Semantic Bridging. It initiates with two distinct inputs: *raw data*, such as transactional logs in CSV or RDF formats, and *business knowledge*, tacit expertise provided by domain specialists. These inputs undergo a process of formalization through *Semantic Onboarding* and *Ontological Unpacking*, respectively. While the former focuses on the instantiation of entities into a foundational ontology, the latter codifies the ”rules of the world” that define how relationships should be interpreted.

The integration of these flows occurs in the *Semantic Bridging* stage. This layer acts as the execution engine where abstract axioms meet concrete instances. By unifying these semantic spaces, the framework moves beyond simple pattern matching, identifying the specific *relators* and *truthmakers* that justify an action; the process culminates in recommendations, where the logical proof generated during the bridging phase serves as the ground truth for the final natural language explanation, effectively mitigating hallucinations and ensuring high-fidelity reasoning.

4.1 UFO-Guided Semantic Onboarding of Raw Data

This stage focuses on the formal ingestion of raw dataset entities. Through Semantic Onboarding, we perform a semantic lift of the data, transforming flat records into a formal Knowledge Base (KB). Following the principles of the Unified Foundational Ontology (UFO), each entity is categorized into foundational types. This ensures that the system recognizes the existential nature of the data, distinguishing between *agents* (users with intentionality) and *functionalcomplexes* (artifacts with functional roles).

To illustrate this process, we introduce a running example centered on a retail organization preparing a strategic promotional sales campaign. The organization’s objective is to maximize overall profitability by identifying (i) product pairs suitable for bundle discounts, (ii) product categories that should be prioritized for discount incentives, and (iii)

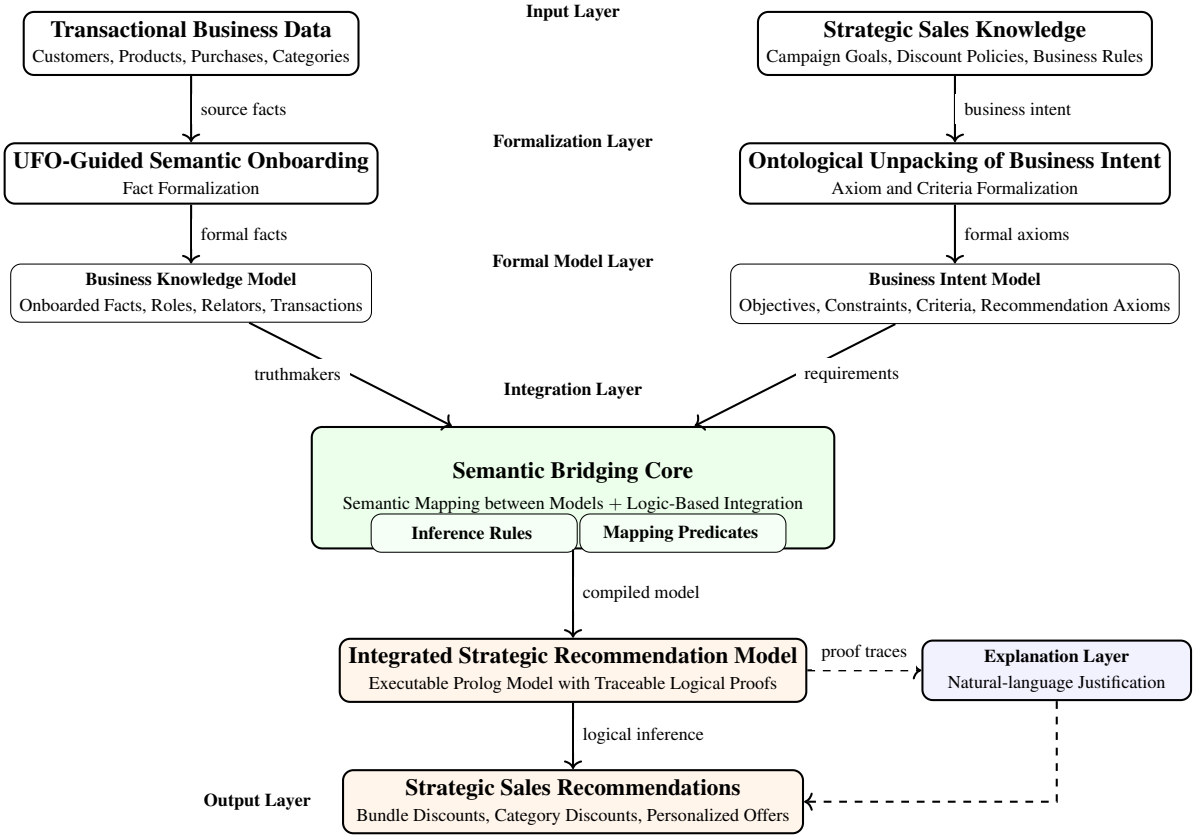


Figure 1: Semantic Bridging architecture for executable strategic recommendation.

personalized product recommendations for specific customers. In this case, the system must reason over explicit business sales criteria in order to determine which promotional actions are most likely to generate commercial gain. In this example, a raw record $\{id : U0534, name : JamesWeber\}$ is transformed into the fact $agent('U0534','JamesWeber',\ldots)$. A product $P102$ is mapped as $functional_complex('P102',\ldots,'B028','C137')$, grounding the data in a formal taxonomy of brands and kinds.

4.2 Ontological Unpacking of Business Intent

The Ontological Unpacking phase defines the "why" of the recommendation by identifying the *truthmakers* that sustain business relationships. In UFO, a relation between two entities is not treated as a mere edge, but as a proposition grounded on the existence of a mediating *relator*. Accordingly, this phase decomposes opaque graph relations such as `also_buy` into formal business conditions that specify when the corresponding relational truthmaker can be said to exist. In our running example, an `also_buy` relation is interpreted as being supported by a *synergy* relator whenever both products share the same brand, thus defining the ontological criteria that make the recommendation valid within the business domain.

A key distinction of this methodology is that the axiomatic foundations of Ontological Unpacking are derived from high-level natural language business specifications. Typically provided by domain experts or business managers, these descriptions encode organizational concepts, category definitions, and strategic constraints. The Unpacking process translates such qualitative knowledge into formal logical rules, transforming informal business intent into a computable structure aligned with the semantic nuances of the domain.

This phase also establishes a direct connection between real-world domain knowledge and the later LLM-assisted explanation process. Because business criteria are formalized as ontology-guided logical rules, modifications in recommendation strategy can be incorporated by updating the natural language specifications that generate the ruleset, without requiring changes in the reasoning architecture. As a

result, domain experts remain responsible for defining the real-world facts that serve as *truthmakers*, ensuring that every generated recommendation remains logically grounded and contextually faithful.

4.3 Semantic Bridging

The final stage, Semantic Bridging, integrates the two previously independent formal spaces produced by Semantic Onboarding and Ontological Unpacking. Its role is to allow the abstract business axioms derived from intent to find their corresponding **truthmakers** in the onboarded business facts. Although both layers are individually formalized, they remain disconnected until explicit semantic correspondences are established between compatible ontological components.

This integration is performed by identifying, for each recommendation requirement, the set of onboarded entities and relations capable of satisfying its ontological conditions. In practical terms, abstract predicates such as *bundle_candidate(X,Y)*, *discount_priority(C)*, or *personalized_offer(U,P)* are transformed into executable predicates whose satisfiability can be tested against concrete transactional evidence. The result of this process is a new formal artifact, called the *Recommendation Model*, which consolidates in a single executable layer: (i) the sales objectives derived from business intent, (ii) the truth-bearing commercial facts extracted from data, and (iii) the bridging rules that determine under which semantic conditions a strategic recommendation proposition becomes valid.

Because the alignment between business predicates and onboarded facts is not assumed to be perfectly complete, the Semantic Bridging engine performs a constrained satisfiability check in which each recommendation rule is decomposed into mandatory and optional conditions. Mandatory predicates must find valid transactional or ontological truthmakers for the recommendation to be generated, whereas optional predicates provide additional semantic support when available. If a mandatory condition fails, the recommendation candidate is discarded; if only optional evidence is absent, the recommendation may still be produced with lower formal completeness, which is later reflected in the Logical Soundness evaluation.

Finally, because the translation of natural-language business strategies into executable Prolog rules is assisted by an LLM, a representative subset of generated axioms is subjected to expert-in-the-loop verification to assess semantic fidelity and rule correctness. This validation serves as a quality assurance mechanism for the ontology-guided unpacking process rather than as a mandatory operational dependency for every generated artifact. Once validated, the resulting rule templates can be ystematically reused across equivalent recommendation scenarios without introducing additional human overhead.

5 Architecture and Implementation

The system architecture is organized into five functional modules that map to the pipeline stages, implemented in Python, leveraging its ecosystem for data manipulation and AI integration, and SWI-Prolog as the core symbolic reasoning engine. The architecture emphasizes the separation between the Data Ingestion layer (Python-based), the Knowledge Modeling layer (Natural Language to Prolog translation supported by LLMs), and the Reasoning Layer (executed within the Prolog environment). The final Explanation Layer acts as the interface, synthesizing the logical output into a human-readable format. Module Description and Interconnections:

The CSV/RDF Ingestion module processes heterogeneous raw data. For this implementation, we developed custom Python parsers that read transactional data from CSV files and transform them into standardized internal representations. The Semantic Onboarding engine executes the mapping between parsed data and the Unified Foundational Ontology (UFO). It is responsible for generating two types of Prolog files: entities.pl, which contains typed individual facts, and triples.pl, which stores the material relations. Table 1 illustrates the Semantic Onboarding process, which transforms raw data into structured, semantically meaningful representations. Rather than treating ingested data as flat records, the approach systematically maps them into ontologically grounded constructs, such as functional complexes and material relations. This transformation is a key contribution of the proposed methodology, as it enables the conversion of raw data sources into a unified, logic-based representation suitable for reasoning. By generating Prolog predicates that explicitly encode entities and their relations, the onboarding process establishes the structural foundation required for subsequent semantic alignment and inference.

The Ontological Unpacking Modeler supports the translation of natural language business rules into formal Prolog axioms (rules.pl). We utilize a Large Language Model (Gemini 2.0) to assist a human ontologist in codifying qualitative descriptions, for example *customers value brand consistency*, into precise

Table 1: Examples of data ingestion and semantic mapping into Prolog.

Stage	Representation
Example 1	

Product ingestion	`product_id, product_name, brand_id, category_id`
	`P102, Samsung 980 SSD, B028, C137`
	`P205, Samsung Odyssey Monitor, B028, C141`

Semantic mapping	`functional_complex('P102','Samsung 980 SSD','B028','C137').` `functional_complex('P205','Samsung Odyssey Monitor','B028','C141').`
Example 2	
Transaction ingestion	`user_id, product_id, timestamp` `U0534, P102, 2023-11-15T10:00:00Z`
Semantic mapping	`buys('U0534','P102','2023-11-15T10:00:00Z').`

Table 2: Example of translating a business strategy into executable Prolog rules.

Stage	Representation
Business Strategy	
Natural-language intent	`Suggest products from the same brand to leverage brand loyalty (Ecosystem Synergy).`
Semantic interpretation	`ecosystem_synergy(P1, P2)`
Executable rule generation	`unpack_relation(P1,P2, ecosystem_synergy) :- functional_complex(P1,_,B,_), functional_complex(P2,_,B,_), P1 \= P2.`

logical clauses that identify the corresponding UFO *Relators*. Table 2 presents the Ontological Unpacking process, in which high-level business strategies expressed in natural language are translated into executable logical rules. This step goes beyond traditional information extraction by identifying underlying relational patterns and mapping them to ontological constructs, such as relators and role-based dependencies. The resulting Prolog rules capture not only the intended behavior of the strategy but also its semantic structure, enabling traceable and explainable reasoning. This constitutes a central contribution of the paper, as it bridges the gap between informal domain knowledge and formal representations that can be directly used in automated decision-making.

The Semantic Bridging process is implemented as a compilation layer of meta-interpreter rules within SWI-Prolog, responsible for integrating the two formal artifacts generated in the previous stages: the *Business Knowledge Model* and the *Business Intent Model*. Rather than querying these models independently, the architecture dynamically loads both sets of predicates and generates an additional executable inference layer composed of mapping predicates, satisfiability predicates, and strategic recommendation rules. This integrated layer materializes the *Integrated Strategic Recommendation Model*, in which high-level commercial objectives are formally connected to concrete transactional truthmakers through explicit logical conditions.

At runtime, the Prolog engine performs deductive unification over this integrated model in order to determine whether a strategic recommendation proposition is logically satisfiable. In practical terms, this means verifying whether the onboarded business evidence jointly satisfies the recommendation axioms extracted from the organizational sales intent. As a result, recommendations are not produced by heuristic ranking or graph traversal, but by formal proof construction. For instance, the inference process may establish that two products satisfy the ontological and transactional conditions required for a bundle discount campaign, or that a given product category fulfills the strategic criteria for promotional discount prioritization. Each recommendation therefore emerges as an executable logical conclusion whose supporting truthmakers remain explicitly identifiable.

To make these formal proofs accessible to decision makers, the architecture includes an LLM-Assisted Explanation and Verification module. This component receives the structured logical derivation produced by the Prolog engine and translates it into a natural-language commercial justification. Importantly, the LLM is not responsible for generating the recommendation itself, but only for verbalizing the already validated proof traces. Prompt constraints require the explanation to remain strictly anchored in the logical evidence supplied by the reasoning layer, ensuring that the final textual justification is a faithful semantic rendering of the underlying formal recommendation proof.

6 Evaluation

To evaluate the proposed approach, we implemented a comparative baseline based on a Knowledge Graph + Large Language Model recommendation architecture inspired by (Seabra et al., 2025). The baseline preserves the same fundamental graph-based recommendation principles while being re-executed under the controlled experimental conditions adopted in this work.

6.1 Experimental Setup

Both the proposed framework and the reimplemented baseline system were evaluated under identical business objectives and using the same controlled dataset, ensuring comparability in terms of available evidence and recommendation tasks. The comparison isolates

Table 3: Representative outputs for the three strategic sales competency questions.

CQ	Strategic Objective	Representative Output
CQ1	Personalized recommendation	Apple Ergonomic Monitor suggested as complementary offer.
Truthmakers & Justification		Previous laptop purchase + ontological Solution Composition relation between acquired and candidate products. The recommendation is justified as a complementary product inferred from prior computing acquisitions and semantic compatibility.
CQ2	Category discount prioritization	Category C199 selected for discount campaign.
Truthmakers & Justification		Category mentioned by user + 3 prior purchases + 6 also-buy targets. The recommendation is justified by high historical affinity and strong downstream cross-selling potential.
CQ3	Bundle discount recommendation	Samsung Ergonomic Accessories + Dell Ultrawide Storage identified as best bundle pair.
Truthmakers & Justification		`also_buy = true` + `also_view = true` + cross-category complementarity. The recommendation is justified as a joint promotional offer supported by strong co-consumption behavior.

differences in reasoning methodology rather than differences in data access. The evaluation considers three competency questions corresponding to representative strategic recommendation tasks: personalized recommendation, category discount prioritization, and bundle discount recommendation. For each competency question, both systems received semantically equivalent task definitions and generated recommendation outputs based on the same organizational evidence.

We utilized a curated synthetic dataset designed to emulate a realistic retail recommendation environment while preserving full experimental controllability and reproducibility. The dataset contains 5,200 entities, including 1,200 users, 3,500 products, 300 brands, and 200 product categories, interconnected through 18,600 relationships. These relationships comprise 5,000 purchase interactions, 2,500 co-purchase (*also_buy*) relations, 2,600 co-view (*also_view*) relations, 1,500 explicit user interest declarations (*mentions*), and the corresponding product-brand and product-category assignments. This structure provides heterogeneous behavioral, semantic, and organizational signals suitable for evaluating recommendation reasoning under controlled conditions. To support reproducibility, the source code and experimental artifacts are publicly available[1], while preserving organizational confidentiality.

6.2 Baseline Approach

The baseline approach models domain knowledge as a Knowledge Graph, where entities are connected through relations such as *also_buy*, *also_view*, and *belongs_to*. Each relation is associated with a weight that influences recommendation probabilities. RDF triples are transformed into textual representations and injected into the LLM prompt, along with domain-specific instructions that guide the recommendation process. While this approach enables the generation of explainable recommendations, relations are treated primarily as weighted edges, and reasoning is guided by prompt engineering rather than formal inference mechanisms.

The selected baseline was chosen because it represents a structurally comparable recommendation architecture previously proposed for the same application domain. In this study, the baseline design was reimplemented over the controlled experimental dataset adopted here, allowing methodological comparison under equivalent evidence conditions.

6.3 Evaluation Strategy and Metrics

Both our proposed framework and the baseline approach are evaluated against the same set of three core Competency Questions (CQs). These questions materialize the business intent and act as the benchmark for our reasoning engine, forcing the system to navigate the ontological structure and derive answers from formal rules. These questions comprise CQ1 (Predictive Reasoning: *What other products a user could buy?*), which evaluates the ability to identify items based on user profile and semantic similarity; CQ2 (Strategic Reasoning: *Which product categories should be prioritized for a discount?*), which assesses reasoning over category-level ontologies and business rules; and CQ3 (Relational Reasoning: *Which two products would be the most suitable candidates for a bundle discount?*), which tests the capacity to identify *relator* dependencies between items using UFO-based modeling.

In fact, the three competency questions activate distinct but complementary layers of strategic

[1] https://anonymous.4open.science/r/semantic_bridging-A4EA

sales reasoning. CQ1 addresses *personalized customer recommendation*, in which the system combines customer-specific transactional history with ontological product compatibility to infer complementary offers, evidencing its ability to generate individualized recommendations through formal product-to-product relational reasoning. CQ2 shifts the focus to *portfolio-level strategic prioritization*, requiring the aggregation of heterogeneous business truthmakers — such as category mentions, prior purchases, and downstream cross-selling opportunities — into a unified decision score for discount targeting, thus illustrating the framework's capacity for multi-criteria commercial inference. CQ3, in turn, exercises a *pairwise relational reasoning* process aimed at bundle discount recommendation, where the system must simultaneously evaluate co-purchase behavior, co-visualization evidence, brand affinity, and cross-category complementarity to determine whether a product pair is strategically suitable for joint promotion. Together, these three scenarios demonstrate that the proposed Semantic Bridging architecture is capable of supporting customer-centered, category-centered, and relational campaign-oriented recommendations within the same executable inference framework.

To evaluate the effectiveness of the proposed framework, we adopted two complementary metrics capturing both formal recommendation validity and practical decision usefulness. The first metric, *Logical Soundness (LS)*, is a quantitative measure defined as the average percentage of mandatory ontological and business axioms satisfied by each generated recommendation with respect to its corresponding competency question. For each recommendation, the Semantic Bridging layer verifies how many of the formal predicates required by the intended strategic sales objective, including transactional truthmakers, ontological mapping conditions, and strategic inference constraints, are effectively satisfied during logical execution. Thus, LS does not simply indicate whether a recommendation can be produced, but measures the degree to which the recommendation is logically complete with respect to the full set of formal business conditions expected for that strategic task.

$$LS(r) = \frac{\left|\left\{c_i \in C_q \mid c_i \text{ is satisfied by } r\right\}\right|}{|C_q|} \times 100$$

For each recommendation (r), let (C_q = c_1, c_2, ..., c_m) be the set of formal conditions required by competency question (q). Each condition corresponds to a Prolog-verifiable axiom, such as the existence of a transactional truthmaker, an ontological mapping predicate, an intent-alignment condition, or a strategic inference constraint. Logical Soundness is defined as:

Table 4: Quantitative and qualitative evaluation results across the three competency questions.

CQ	Axioms	Axioms Sat./Tot.	LS (%)	ER Base.	ER Prop.
CQ1: Personalized Rec.	6	5.00 / 6.00	83.33	3.4	4.6
CQ2: Category Discount	1	4.00 / 7.00	57.14	3.1	4.3
CQ3: Bundle Suitability	20	5.55 / 7.00	79.29	2.9	4.1
Overall Mean	**27**	**5.37 / 6.78**	**79.37**	**3.1**	**4.3**

The LS score for each competency question is the average LS across its generated recommendations. This metric is objective because each condition is checked against formal predicates, row-level evidence, or Prolog execution results, while still allowing partially supported recommendations to receive intermediate scores.

The second metric, *Expert Rating (ER)*, is a qualitative assessment performed by domain evaluators using a 1–5 Likert scale, considering three dimensions simultaneously: strategic relevance of the recommendation, commercial actionability, and clarity of the generated explanation. While LS captures the internal formal adequacy of the recommendation with respect to the intended business axioms, ER captures how useful and convincing the resulting recommendation appears from a practical decision-making perspective.

The evaluation results reveal distinct formal and qualitative behaviors across the three competency questions. From a quantitative perspective, the *Logical Soundness* values indicate that the generated recommendations satisfy, on average, a substantial portion of the mandatory ontological and business axioms encoded for each strategic sales task. CQ1 achieved the highest LS (83.33%), showing that personalized recommendation is strongly supported when customer purchase history can be consistently connected to semantic product compatibility relations. CQ3 also presented a high LS (79.29%), indicating that most bundle discount candidates satisfy the expected combination of co-purchase evidence, relational complementarity, and strategic bundling conditions. In contrast, CQ2 obtained the lowest LS (57.14%), suggesting that category discount prioritization depends on a broader set of aggregate business criteria, some of which were only partially verifiable through the current formal predicates. Overall, the mean LS of 79.37% indicates that the Semantic Bridging layer is generally able to produce recommendations that are formally close to the intended strategic business conditions.

A similar pattern can be observed in the qualitative assessment. The *Expert Rating* assigned to the pro-

Table 5: Scalability results of the Semantic Bridging inference engine under increasing fact volumes.

Approx. Facts	Load Time (ms)	CQ1 (ms)	CQ2 (ms)	CQ3 (ms)	Mean Query Time (ms)
8,005	1,291.62	3.71	7.82	5.29	5.61
40,005	6,728.19	4.28	0.89	3.97	3.05
80,005	47,180.15	3.22	1.15	2.94	2.44
200,005	213,887.13	6.35	3.02	4.31	4.56
400,005	1,062,286.39	11.39	7.17	6.22	8.26

posed framework remained consistently high across the three scenarios, with an overall mean of 4.3, indicating that the generated recommendations were perceived as strategically coherent, commercially actionable, and accompanied by sufficiently clear justifications. Personalized recommendation again received the highest qualitative evaluation, followed by category discount prioritization and bundle discount suitability. Interestingly, even in CQ2, where Logical Soundness was lower, the expert assessment remained positive, suggesting that some commercially meaningful business signals were recognized by evaluators despite not being fully captured by the strict formal verification process. When compared to the baseline ratings, the proposed recommendations were generally perceived as more convincing from a strategic sales perspective, reinforcing that the contribution of the framework lies not only in formal semantic grounding, but also in producing recommendations that better support practical promotional decision-making.

6.4 Scalability Assessment

In addition to recommendation quality, we evaluated the computational scalability of the proposed Semantic Bridging engine by measuring the behavior of the SWI-Prolog inference layer under progressively larger synthetic transactional knowledge bases. Following prior evidence (Wielemaker et al., 2012) that SWI-Prolog provides a robust and scalable environment for knowledge-intensive application, we generated five datasets ranging from approximately 8 thousand to 400 thousand onboarded facts while preserving the same recommendation rules and competency queries. For each configuration, we measured both the knowledge base loading time and the average inference time required to execute CQ1, CQ2, and CQ3.

The results indicate that, although the one-time loading cost of the Prolog fact base increases substantially with the number of onboarded artifacts, the runtime recommendation inference remains highly stable. Even at the largest evaluated configuration, containing more than 400 thousand facts, all three competency questions were answered in less than 12 milliseconds on average. This behavior suggests that the

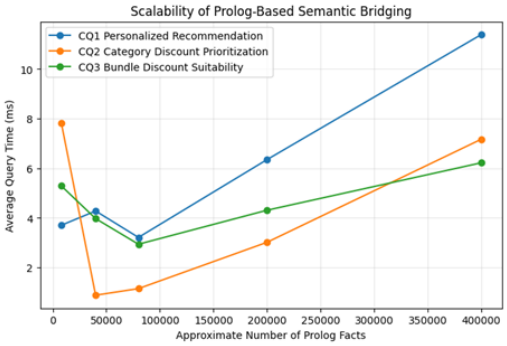


Figure 2: Growth of inference time across the three competency questions as the fact base increases.

indexing and unification mechanisms of SWI-Prolog are able to preserve efficient access to the relevant transactional truthmakers once the knowledge base is materialized. In practical terms, these findings indicate that the proposed semantic recommendation architecture is computationally viable for large business repositories, with the main scalability cost concentrated in fact ingestion rather than in logical recommendation generation itself.

7 Conclusions and Future Work

This paper presented an ontology-guided recommendation framework designed to bridge a critical gap in strategic sales recommendation systems: the semantic misalignment between high-level business intent and the transactional evidence available in organizational data. By explicitly formalizing both the strategic commercial objectives and the underlying business facts, the proposed approach enables recommendation generation to be treated as a formal satisfiability problem rather than a mere association retrieval task. The Semantic Bridging layer plays a central role in this process by integrating the *Business Intent Model* and the *Business Knowledge Model* into a unified executable inference structure. As demonstrated in the evaluation, this alignment allows the system to produce recommendations that are not only commercially plausible, but also semantically coherent with the intended sales objective.

Beyond the specific recommendation scenarios

explored in this work, the proposed framework contributes a broader conceptual shift in how recommendation systems can be understood in strategic business environments. Rather than treating recommendations as outputs derived from statistical proximity, graph traversal, or behavioral association patterns alone, the proposed approach frames recommendation generation as a semantically constrained decision-support process grounded in explicit business semantics. This distinction is particularly relevant in enterprise contexts, where recommendations must often satisfy formal organizational criteria, commercial objectives, and explainability requirements that cannot be fully captured through correlation-driven models alone. In this sense, the contribution of this work extends beyond recommendation accuracy, introducing a reusable neuro-symbolic decision framework for semantically justified commercial inference.

Another relevant outcome of this work is the demonstration that recommendation explainability can be grounded on explicit logical reasoning over truthmakers. Since each recommendation is derived through deductive inference on formally represented business relations, products, and customer evidence, the resulting explanations are traceable to concrete logical proofs rather than post-hoc textual rationalizations. This reinforces the potential of ontology-guided executable reasoning not only as a mechanism for recommendation generation, but also as a principled foundation for trustworthy explainable AI in decision-support applications.

This opens a promising avenue for future research in at least three directions: first, scaling the framework to larger and more heterogeneous retail environments; second, extending the formal business intent model to support more complex commercial strategies such as dynamic pricing, inventory-aware campaigns, and adaptive promotional orchestration; and third, investigating the use of richer ontological business semantics to support multi-objective recommendation scenarios involving profitability, customer retention, customer lifetime value, and long-term promotional optimization.

REFERENCES

Gao, C., Wang, X., He, X., and Li, Y. (2022). Graph neural networks for recommender system. In *Proceedings of the fifteenth ACM international conference on web search and data mining*, pages 1623–1625.

Guarino, N. and Guizzardi, G. (2015). We need to discuss the truthmakers of relational propositions. In *Proceedings of the 7th International Conference on Formal Ontology in Information Systems (FOIS 2015)*, pages 17–30. IOS Press.

Guizzardi, G. (2005). *Ontological foundations for structural conceptual models*. Telematica Instituut, Enschede, The Netherlands.

Guizzardi, G. (2008). On the ontological foundations of conceptual modeling: Roles and relations.

Guizzardi, G. and Wagner, G. (2010). Towards ontological foundations for conceptual modeling languages. In *Theory and Applications of Ontology: Computer Applications*, pages 37–63. Springer.

Guizzardi, G. and Wagner, G. (2012). Ontological analysis of relationships in conceptual modeling. *Data & Knowledge Engineering*, 77:1–30.

Guizzardi, G., Wagner, G., Almeida, J. P. A., and Guizzardi, R. S. S. (2015). Towards ontological foundations for conceptual modeling: The unified foundational ontology (ufo) story. *Applied Ontology*, 10(3-4):259–271.

Huang, Y., Chen, X., Nie, L., and Zhang, Y. (2022). Knowledge-aware recommendation: A survey of knowledge graphs and beyond. *ACM Computing Surveys*, 55(11):1–37.

Lin, J., Dai, X., Xi, Y., Liu, W., Chen, B., Zhang, H., Liu, Y., Wu, C., Li, X., Zhu, C., et al. (2025). How can recommender systems benefit from large language models: A survey. *ACM Transactions on Information Systems*, 43(2):1–47.

Lowe, E. J. (2001). *The Possibility of Metaphysics: Substance, Identity, and Time*. Oxford University Press, Oxford.

Ma, T., Zhou, X., Wang, C., and Fang, H. (2021). Neusymrec: A neuro-symbolic recommendation framework via logical rule learning. In *Proceedings of The Web Conference*.

Ren, X., Chen, T., Nguyen, Q. V. H., Cui, L., Huang, Z., and Yin, H. (2023). Explicit knowledge graph reasoning for conversational recommendation. *arXiv preprint arXiv:2305.00783*.

Seabra, A., Cavalcante, C., and Lifschitz, S. (2025). Semantic prompting over knowledge graphs for next-generation recommender systems. In Aberer, K., Domingo-Ferrer, J., and Marchiori, M., editors, *Proceedings of the 21st International Conference on Web Information Systems and Technologies, WEBIST 2025, Marbella, Spain, October 21-23, 2025*, pages 394–403. SCITEPRESS.

Strawson, P. F. (1959). *Individuals: An Essay in Descriptive Metaphysics*. Methuen, London.

Tiwary, N. et al. (2024). A review of explainable recommender systems utilizing knowledge graphs and reinforcement learning. *IEEE Access*.

Wang, X., He, X., Wang, M., Feng, F., and Chua, T.-S. (2019). Kgat: Knowledge graph attention network for recommendation. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 950–958. ACM.

Wielemaker, J., Schrijvers, T., Triska, M., and Lager, T. (2012). Swi-prolog. *Theory and Practice of Logic Programming*, 12(1-2):67–96.

Xie, L., Hu, Z., Cai, X., Zhang, W., and Chen, J. (2021). Explainable recommendation based on knowledge graph and multi-objective optimization. *Complex & Intelligent Systems*.

Yang, Y., Huang, C., Xia, L., and Huang, C. (2023). Knowledge graph self-supervised rationalization for recommendation. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, KDD '23, pages 2775–2784. ACM.